Demo Recording Data & Test Scenarios

As built and tested on 31 October 2026

Deliverable 5 of 5. These are the values you type during the key-features demo video.

The sequence matches the Key Features User Manual and the Demo Video Script, scene for scene. Every field, option and rule here was checked against the application's forms and database on 31 Oct 2026 (database `lms_dev`, migration 0059).

> All people, CNICs, IBANs, phone numbers and parcel numbers below are **fictional**.

0. Before you record

#	Check	How
1	The server runs the latest code	`npm run build`, then restart the server on port 3040. The *Create acquisition* button, the drawing tools and the latest fixes need this.
2	Demo sign-in list is visible	Demo mode shows each account and its password on the sign-in page. Record on `localhost`, and **turn demo mode off after recording** (audit item O1).
3	Map tiles load	The map needs internet (OpenStreetMap). Without it, choose Basemap **None**: drawing still works.
4	Browser	Chrome, 1440 × 900 or larger, zoom 100%. Use one **incognito window per user**, or sign out between users.
5	Re-authentication	Releasing a cheque asks for the password again if the sign-in is older than 10 minutes. Keep `mland`'s password at hand.
6	Codes	The codes in "Expected result" (e.g. `REQ-2026-0004`) assume nothing else is created first. If a number differs, carry on: only the number changes.
7	Retake	If you need a second take, use the **Take-2 values** in §12 so you do not hit "already exists" errors.

The cast (demo accounts)

Username	Person (fictional)	Role / approval position	Province	Appears in scene
`lofield`	Sanaullah Khoso	Land Officer (Field)	Sindh	1, 2, 3, 6, 7, 8
`explorer`	Usman Ghani	Exploration Planner	Sindh	4, 5
`dirops`	Javed Akhtar	Director Operations	all	5
`regmgr`	Sadia Memon	Regional Manager (Sindh)	Sindh	5

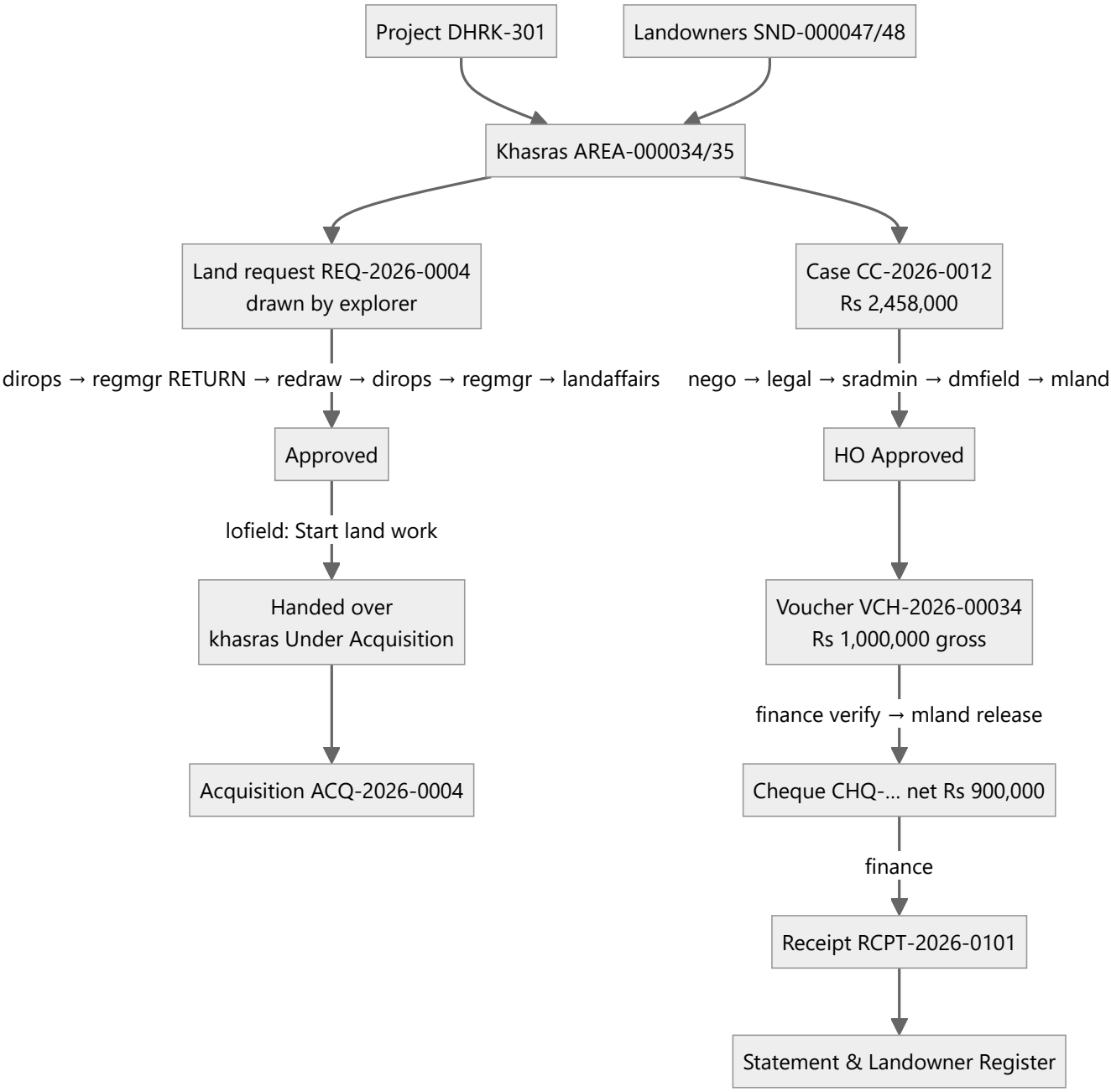
Username	Person (fictional)	Role / approval position	Province	Appears in scene
`landaffairs`	Zubair Soomro	DM / TL Land Affairs	Sindh	5
`nego`	Bilal Chandio	Negotiation Officer	Sindh	7
`legal`	Ayesha Siddiqui	Legal Advisor	all	7
`sradmin`	Faisal Shaikh	Sr. Administrator LMS (Field)	Sindh	7
`dmfield`	Nadia Baig	DM / Team Lead (Field)	Sindh	7
`mland`	Imran Qadir	Manager Land (HO)	all	7, 8, 9
`finance`	Shazia Noor	Finance Officer	all	8, 9

---

1. The story

Daharki Well-03 is a new exploratory well near Jarwar, District Ghotki, Sindh.

1. The **Land Officer** registers the project, two khasras and their owners (two brothers).
2. The **Exploration Planner** marks the land needed on the map.
3. Three approvers decide; the Regional Manager **returns it once** to move the boundary.
4. The Land Officer takes it up, the khasras become *Under Acquisition*, and an **acquisition** opens pre-filled.
5. A **compensation case** values lease, crop, trees and disturbance. Five positions approve it.
6. **Finance and the Manager Land** pay the first owner: WHT is deducted automatically, a cheque is released, and a receipt is printed.
7. The **reports** show the owner's statement and the landowner register.



2. Scene 1: Project (user `lofield`)

Projects / Well Locations → + Add Project

Field	Value	Required
Project code	`DHRK-301`	Yes (2–32 letters, digits, `-` `/` ; unique)
Province	`Sindh` (SND)	Yes
Name	`Daharki Well-03 Exploratory Location`	Yes
Type	`Well Location`	Yes

Field	Value	Required
Block / concession	`Daharki D&PL`	Yes
District	`Ghotki`	Yes
Required area (acres)	`20`	Optional
Description	`Exploratory well pad, rig camp and 0.8 km access road north of Jarwar. Spud planned Feb 2027.`	Optional

Expected result: the project **DHRK-301** is listed with status **Active**.

The project form has no map position. The project's location comes from its khasras and its land request.

---

3. Scene 2: Landowners and khasras (user `lofield`)

3a. Landowner 1: **Landowners → + Add Landowner**

Field	Value	Required
Province	`Sindh`	Yes
CNIC	`45104-2000011-3` (type the 13 digits; dashes appear by themselves)	Yes (unique)
Full name (as on CNIC)	`Ghulam Rasool Mahar`	Yes
Father / husband name	`Muhammad Bux Mahar`	Yes
Owner type	`O — Owner`	Yes
Tax status	`Filer`	Yes (Filer → 10% WHT)
Phone	`0301-2345678`	Optional
Bank branch	`HBL Daharki`	Optional
IBAN	`PK24HABB0012345678901234`	Optional (24 characters, format checked)
Address	`Village Jarwar, Taluka Daharki, District Ghotki`	Optional

Expected result: SND-000047, Active.

3b. Landowner 2: **+ Add Landowner**

Field	Value
Province	`Sindh`

Field	Value
CNIC	`45104-2000012-1`
Full name	`Nazir Ahmed Mahar`
Father / husband name	`Muhammad Bux Mahar`
Owner type	`O — Owner`
Tax status	`Non-Filer` (→ 20% WHT, shown in scene 8 option B)
Phone	`0302-3456789`
Bank branch	`Meezan Bank Daharki`
IBAN	`PK57MEZN0003450123456789`
Address	`Village Jarwar, Taluka Daharki, District Ghotki`

Expected result: SND-000048, Active.

3c. Khasra 1: **Survey / Khasra Register → + Add Survey**

Field	Value	Required
Province	`Sindh`	Yes
Revenue location	`Ghotki → Daharki → Jarwar`	Yes
Project	`DHRK-301 — Daharki Well-03 Exploratory Location`	Optional (links the khasra to the project)
Survey / khasra no.	`451/2`	Yes
Block no.	`9`	Optional
Khasra type	`Rural`	Optional
Land class	`Private`	Yes
Recorded area	`12` with unit `Acre / Killa`	Yes
Latitude	`28.0710`	Optional (needed for the map)
Longitude	`69.7050`	Optional (give both or neither)
Remarks	`Cultivated; tube-well on the east side.`	Optional

Expected result: AREA-000034, status **Proposed**, area 12.00 acres.

3d. Khasra 2: **+ Add Survey**

Field	Value
Province	`Sindh`
Revenue location	`Ghotki → Daharki → Jarwar`

Field	Value
Project	`DHRK-301`
Survey / khasra no.	`452/1`
Block no.	`9`
Khasra type	`Rural`
Land class	`Private`
Recorded area	`64` with unit `Kanal` (shows unit conversion: 64 Kanal = **8.00 acres**)
Latitude	`28.0703`
Longitude	`69.7062`
Remarks	`Wheat crop standing at the time of survey.`

Expected result: AREA-000035, Proposed, 8.00 acres.

3e. Owner shares

In the **Survey / Khasra Register**, on the khasra's row, click **Shares**. Add each owner with "Landowner to add" and **+**, then **Save shares**. The total must be exactly 100%.

Khasra	Owner	Share %	Revenue document
AREA-000034	Ghulam Rasool Mahar (SND-000047)	`60`	`Fard 2026/118`
AREA-000034	Nazir Ahmed Mahar (SND-000048)	`40`	`Fard 2026/118`
AREA-000035	Ghulam Rasool Mahar	`50`	`Fard 2026/119`
AREA-000035	Nazir Ahmed Mahar	`50`	`Fard 2026/119`

Expected result: the shares column shows **100.0%** and **2 owner(s)**.

Negative demo (optional): enter 60 + 30. The **Save shares** button stays disabled until the total is 100%.

---

4. Scene 3: The khasras on the GIS map (user `lofield`, optional)

GIS Land Map: the two new markers appear near Jarwar, grey (*Proposed*). Click one to see its code, status, area and location.

---

5. Scene 4: Exploration Planner draws the land (user `explorer`)

Sign in as `explorer`. The planner lands **directly on the GIS Land Map**: this role has no dashboard, and no Landowners or Payments menu.

1. Click ✎ **Draw land request**.
2. Click **Paste lat/long** and paste the four corners below: `` `28.07200, 69.70400 28.07200, 69.70700 28.06950, 69.70700 28.06950, 69.70400` ``
3. Click **Use these corners**.
4. *(Optional)* Click once on the map to add a fifth corner, then remove it with ×. This shows that corners can be clicked, dragged and removed.
5. Click **Continue: request details**.

Expected panel figures:

Item	Value
Corners	4
Area	**20.19 acres · 8.17 ha**
Centre	28.07075, 69.70550
Parcels inside	**2: AREA-000034, AREA-000035**

Land request details form:

Field	Value	Required
Project	`DHRK-301 — Daharki Well-03 Exploratory Location (SND)`	Yes
Purpose	`Well Location`	Yes
Priority	`Urgent`	Default Standard
Required by	`2027-01-15`	Yes
Target deh / village	`Jarwar` (pre-filled from the parcels inside)	Optional
Concession block	`Daharki D&PL`	Optional (blank = Mari Concession)
Justification	`Exploratory well Daharki Well-03: well pad 120 m × 120 m, rig camp and mud pits, plus a 0.8 km access road. Rig mobilisation is planned for Feb 2027, so the land is needed by 15 Jan 2027.`	Needed to submit

Click **Submit to Director Operations**. *(Alternative: Save as draft, then submit from Land Requests → Send.)*

Expected result:

- The message "Created **REQ-2026-0004** and sent it to Director Operations."
- The polygon turns **orange** (*Under review*).
- **Land Requests** shows the workflow at step 2 and "Waiting on Director Operations".

---

6. Scene 5: Three-tier approval with one return

Every decision needs an **Audit comment**.

Step	Sign in as	Where	Action	Comment to type	Expected stage after
5.1	`dirops`	Land Requests → REQ-2026-0004 → **Details**, then **Map**, then **Review**	Approve	`Operational need confirmed: Well-03 is in the 2027 drilling plan.`	Under Review - Regional
5.2	`regmgr`	Land Requests → **Review**	**Return**	`Pull the north boundary about 45 m south to keep clear of the Jarwar link road, then resubmit.`	**Returned** (waiting on the initiator)
5.3	`explorer`	Land Requests → **Edit on map**	See below	—	—
5.4	`dirops`	Review	Approve	`Revised boundary acceptable.`	Under Review - Regional
5.5	`regmgr`	Review	Approve	`Region supports it; no conflict with other activity.`	Under Review - Land Affairs
5.6	`landaffairs`	Review	Approve	`Private land, two owners identified, no litigation recorded. Approved.`	**Approved** (green)

Step 5.3, the redraw by `explorer`:

1. Click **Clear**, then **Paste lat/long**, and paste the revised corners below. Then **Use these corners**. `` `28.07160, 69.70400 28.07160, 69.70680 28.06950, 69.70680 28.06950, 69.70400` `` Expected: **15.83 acres · 6.41 ha**, still 2 parcels inside.
2. Click **Continue: review and save**.
3. In **What changed**, type: `North boundary moved 45 m south and east edge 20 m west, as asked by the Regional Manager.`
4. Click **Save and send to Director Operations**.

Expected: the stage goes back to **Submitted**, and **Details** → **Polygon versions** lists **v2 (current)** and **v1**.

Negative checks you can show:

- `regmgr` gets no **Review** button while the request waits for Director Operations.
- `explorer` cannot approve their own request.
- In 5.4, `dirops` decides again: after a return, every tier decides afresh.

---

7. Scene 6: Hand-over and acquisition (user `lofield`)

1. **Land Requests** → REQ-2026-0004 → **Start land work**.

Field	Value
Land work	`Acquisition`

Field	Value
Mark the recorded parcels inside the polygon "Under Acquisition"	✓ (ticked)
Comment	`Negotiation with the Mahar family starts 10 Nov 2026.`

Click **Take it up**.

Expected result:

- "REQ-2026-0004 handed over for acquisition. 2 parcel(s) marked Under Acquisition: AREA-000034, AREA-000035."
- The stage is **Handed over** (blue on the map).
- Both khasras are now **Under Acquisition** in the Survey register.

2. On the same row, click **Create acquisition**. The form **"Add Acquisition — from REQ-2026-0004"** opens pre-filled:

Field	Pre-filled / enter	Required
Project	DHRK-301 (pre-filled)	Yes
Mode	`Private Lease`	Yes
Area required	15.83 (pre-filled from the polygon)	Yes
Negotiated rate (Rs / acre)	`60000`	Optional
Target handover	2027-01-15 (pre-filled)	Optional
Surveys	AREA-000034, AREA-000035 (pre-filled chips)	Optional
Remarks	`From land request REQ-2026-0004 (Well Location).` (pre-filled)	Optional

Click **Save**.

Expected result:

- **ACQ-2026-0004** is created at stage **Identification**.
- Land Requests → Details shows "record **ACQ-2026-0004**".

3. *(Optional)* Click **Title** and tick:
    - Fard / Record of Rights
    - Latest mutation verified
    - CNIC of every owner verified

Each tick records who and when.

---

8. Scene 7: Compensation case and its approval chain

8a. Create the case (user `lofield`)

Compensation Cases → + Add Compensation Case

Header:

Field	Value	Required
Project / well location	`DHRK-301`	Yes
Land used for	`Well Location`	Yes
Lease rate (Rs / acre / year)	`60000`	Required when there are lease lines
Lease term (months)	`36`	Required when there are lease lines
Remarks	`Three-year surface lease for the Well-03 pad, with crop and tree damage.`	Optional

Landowners (+ Add line). The shares must total 100%.

Landowner	Share %	As
Ghulam Rasool Mahar (SND-000047)	`60`	`O — Owner`
Nazir Ahmed Mahar (SND-000048)	`40`	`O — Owner`

Lease lines

Survey	Owners' share %	Area (ac)	Reduced area (ac)
AREA-000034	`100`	`12`	*(blank)*

Crop compensation

Survey	Crop	Rate (Rs/ac)	Damaged (ac)
AREA-000035	`Wheat`	`45000`	`5`

Tree compensation

Survey	Species	Size	Rate (Rs)	Count
AREA-000035	`Kikar`	`Medium`	`8000`	`6`

Other compensation

Survey	For	Amount (Rs)
AREA-000034	`Disturbance allowance`	`25000`

Documents seen: tick all six:

1. CNIC copies of all landowners
2. Fard / record of rights
3. Signed consent / agreement
4. Bank details or IBAN certificate
5. Site photographs
6. Joint survey / measurement report

Click **Create case**.

Expected amounts (calculated by the database, not typed):

Head	Calculation	Amount (Rs)
Lease	60,000 × 12 ac × 36 ÷ 12	2,160,000
Crop	45,000 × 5 ac	225,000
Trees	8,000 × 6	48,000
Other	—	25,000
Total award		**2,458,000**

Expected result:

- **CC-2026-0012**, stage **Prepared**.
- `lofield` sees **no approve button**: the preparer cannot approve.

8b. Approval chain: one person, one step

Open the case and use the workflow button. A note is optional, but say it aloud.

Step	Sign in as	Button	Note	Stage after
1	`nego`	→ Checked	`Rates agreed with both owners on site.`	Checked
2	`legal`	→ Reviewed	`Title clear per Fard 2026/118-119; no litigation.`	Reviewed
3	`sradmin`	→ Verified	`Field records and measurements verified.`	Verified
4	`dmfield`	→ Field Approved	`Approved at field level.`	Field Approved
5	`mland`	→ HO Approved	`Approved at head office.`	**HO Approved** (locked)

Negative checks:

- `nego` cannot approve two steps; the second attempt is refused.
- Step 4 is refused if a required document is unticked, with "Missing documents: …".

---

9. Scene 8: Payment, WHT, cheque and receipt

9a. Request the voucher (user `lofield`)

Payments & Cheques → + Request Payment

Field	Value	Required
Project	`DHRK-301`	Yes
Case	`CC-2026-0012 · Rs 2,458,000 outstanding`	Optional (only HO Approved cases are listed)
Payee (landowner)	`Ghulam Rasool Mahar — Filer`	Required when a case is chosen
Category	`Lease Rent`	Yes
Pay type	`Advance Lease (multi-year)`	Yes
Payment mode	`Cross Cheque`	Yes
Land class	`Private`	Yes
Period from	`2026-11-01`	Yes
Period to	`2029-10-31`	Optional (on or after the start)
Gross amount (Rs)	`1000000`	Yes
Remarks	`First instalment: advance lease rent for the Well-03 pad (60% owner).`	Optional

Expected result:

- **VCH-2026-00034**, stage **Requested**.
- WHT **10%** = Rs 100,000; net **Rs 900,000**. The rate is set by the system from the owner's Filer status.

Option B (shows the non-filer rate): a second voucher for **Nazir Ahmed Mahar**: category `Crop Compensation`, pay type `One-time`, mode `Direct Credit (IBFT)`, gross `500000`.

- Expected WHT is **20%** = Rs 100,000; net Rs 400,000.

Negative checks:

- Gross `2500000`: refused, "case CC-2026-0012 award is 2458000.00 … at most … can still be requested".
- A typed payee name instead of an owner: refused.

9b. Verify (user `finance`)

Payments → Vouchers → VCH-2026-00034 → **Verify**. Note: `Payee, WHT and case checked.`

Expected result: stage **Verified**. The budget check runs here.

9c. Release the cheque (user `mland`)

Payments → VCH-2026-00034 → **Release cheque**.

Field	Value
Cheque date	`2026-11-05` (or today)
Note	`Released to the owner at Daharki field office.`

If asked, re-enter `mland`'s password.

Expected result:

- Stage **Released**; cheque **CHQ-25186412** for **Rs 900,000** (net).
- A SAP posting is queued.

9d. Record and print the receipt (user `finance`)

Payments → **Cheques** tab → CHQ-25186412 → **Record receipt**.

Field	Value	Required
Receipt no.	`RCPT-2026-0101`	Yes
Received on	`2026-11-05`	Yes
Acknowledged by	`Ghulam Rasool Mahar`	Yes
Their CNIC	`45104-2000011-3`	Optional
Remarks	`Cheque received in person.`	Optional

Then click **Print receipt** in the Receipts table.

Expected result: a printable *Receipt of Payment* showing:

- the parcel (Survey 451/2, Deh Jarwar, Taluka Daharki, District Ghotki)
- the heads
- Gross 1,000,000; less WHT (Filer) 100,000; **Total 900,000**
- the amount in words
- the settlement (payable to this owner, paid before, this payment, balance)
- the declaration and signature blocks

9e. Allocate to the case (user `lofield` or `finance`, optional)

Payments → **Liabilities** tab → CC-2026-0012 → **Allocate voucher**:

- Voucher: `VCH-2026-00034`
- Amount: `900000` (the net amount)

Expected result: the case shows **Part paid**, and Ghulam Rasool Mahar's statement balance falls by the allocated amount.

---

10. Scene 9: Reports and dashboard

Step	Sign in as	Where	What to show	Expected
10.1	`lofield`	**Landowners** → SND-000047 → **Statement**	Owed, paid, balance; payment lines gross with WHT in the description	The award share and Rs 1,000,000 paid (900,000 cheque + 100,000 WHT)
10.2	`mland`	**Records & Reports** → **Landowner Register**	Province `Sindh`, District `Ghotki`, Taluka `Daharki`, Deh `Jarwar`; Area unit `All in Kanal`; Show `One row per owner`	Ghulam Rasool Mahar and Nazir Ahmed Mahar with their khasras and shares; 12 ac = **96 Kanal**, 8 ac = **64 Kanal**
10.3	`mland`	Export buttons **PDF / Excel / CSV**	Download	The file has the same rows as the screen
10.4	`mland`	**Executive Dashboard**	Province chart, paid vs approved, approvals waiting	Figures include the new case and cheque
10.5	`audit`	**Audit Trail**	Filter by record `REQ-2026-0004` or `CC-2026-0012`	Every change, with who and when

11. Validation scenarios (quick "what if" shots)

#	Scenario	Where / who	Input	Expected message / result
V1	Boundary crosses itself	GIS draw (`explorer`)	Paste `28.0720, 69.7040` / `28.0695, 69.7070` / `28.0720, 69.7070` / `28.0695, 69.7040`	"The boundary crosses itself..."; Continue disabled
V2	Corner outside Pakistan	GIS draw	`50.0, 10.0` as one corner	"Every corner must be inside Pakistan..."
V3	Shares ≠ 100%	Shares dialog	60 + 30	Save disabled
V4	Duplicate CNIC	Add Landowner	`45104-2000011-3` again	"A landowner with this CNIC is already registered"
V5	Preparer approves own case	`lofield` on CC-2026-0012	—	No approve button
V6	Over-award voucher	Request Payment	Gross `2500000` on CC-2026-0012	Refused; "at most ... can still be requested"
V7	Province isolation	`loquetta` (Balochistan)	Open Land Requests	REQ-2026-0004 is not listed
V8	Planner sees no owner data	`explorer`	Menu	No Landowners / Payments / Cases; the Survey register has no Shares button

12. Take-2 values (for a second recording)

Use these if the first take's records already exist. All rules are the same.

Item	Take-2 value
Project code / name	`DHRK-302` / `Daharki Well-04 Exploratory Location`
Landowner 1	CNIC `45104-2000021-7`, `Abdul Sattar Lund`, father `Haji Karim Bux Lund`, Filer, IBAN `PK11UNIL0109000300123456`
Landowner 2	CNIC `45104-2000022-5`, `Shabbir Ahmed Lund`, father `Haji Karim Bux Lund`, Non-Filer
Khasra 1	`461/3`, block 10, 12 Acre, lat `28.0810` lng `69.7350`
Khasra 2	`462/1`, block 10, 64 Kanal, lat `28.0803` lng `69.7362`
Polygon v1	`28.08200, 69.73400` / `28.08200, 69.73700` / `28.07950, 69.73700` / `28.07950, 69.73400`
Polygon v2	`28.08160, 69.73400` / `28.08160, 69.73680` / `28.07950, 69.73680` / `28.07950, 69.73400`
Receipt no.	`RCPT-2026-0102`

Areas are about the same as take 1. These coordinates are clear of every existing request and of the Disputed parcel AREA-000011 (28.090, 69.720).